**AI TEXT DETECTED (%)**
Percentage of content showing AI-generated patterns

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**Matched 0 words showing AI-pattern signal**

**POSITION ON SCALE**
Where this score falls on a 0 - 100% range




## 01 Submission Information

Document metadata

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## 02 Content Composition

Colors match in-document highlighting



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SCAN

**AI INTEGRITY HELP**
Guides, policy, and best practices.

► **AI Help Centre**
drillbitglobal.com/ai-content-detection

► **Detection Methodology**
drillbitglobal.com/ai-method

## 03 AI Content Analysis — Block Level

0 Block(s) Analysed

**-- No AI-matched blocks --**

## 04 How to interpret this report

Our methodology, what affects results, and best practices

**HOW DETECTION WORKS**

DrillBit combines established AI-detection methods - including perplexity analysis, burstiness measurement, and stylometric profiling - with proprietary techniques developed in-house. Our engine evaluates submissions across four additional analytical layers: the hierarchy of writing, covering organisation and progression of ideas; the composition pattern, covering sentence structure, transitions, and recurring constructions; word prioritisation, covering the order, weight, and placement of key terms; and the technical composition of the document, covering formatting consistency and structural uniformity. This multi-layered approach yields a more complete and reliable assessment than single-signal detection.

**WHY SCORES VARY ACROSS TOOLS**

Detection systems differ in features, training data, and scoring rules; variation across tools is therefore expected. DrillBit applies a broader signal set and deeper structural analysis than conventional detectors, producing results calibrated for accuracy and consistency. Certain writing styles - formal, translated, or template-based text - may produce elevated signals across any system; in DrillBit's framework, such results reflect the genuine statistical profile of the text rather than methodological noise.

**HOW TO USE THIS REPORT**

This report is designed to give reviewers a dependable evidentiary foundation for assessing authorship. It identifies sections warranting closer examination and presents the analytical basis for each, equipping reviewers, mentors, and academic guides to reach well-informed conclusions. Final decisions rest with the reviewer, considered alongside drafts, prior work, and supporting material - supported by the strength and clarity of DrillBit's findings.

CyberGuard: Privacy-Aware Cybercrime Complaint Classification Using PII Masking and TF-IDF–Logistic Regression Dr. Sadananda L Mohith Raj K Kotian Nikhil G Shetty Associate Professor & Head USN: 4MW23CS076 USN: 4MW23CS082 Department of Computer Science and Department of Computer Science and Department of Computer Science and Engineering Engineering Engineering Shri Madhwa Vadiraja Institute of Technology Shri Madhwa Vadiraja Institute of Technology Shri Madhwa Vadiraja Institute of Technology and Management and Management and Management Bantakal, India Bantakal, India Bantakal, India sadananda.1@gmail.com kotianmohith907@gmail.com nikhilgshetty100@gmail.com Pradyumna Shettigar Pruthvi Shetty USN: 4MW23CS093 USN: 4MW23CS108 Department of Computer Science and Department of Computer Science and Engineering Engineering Shri Madhwa Vadiraja Institute of Technology Shri Madhwa Vadiraja Institute of Technology and Management and Management Bantakal, India Bantakal, India pradyumnashettigar26@gmail.com pruthvishetty04@gmail.com Abstract— CyberGuard is a complaint-intake and triage prototype services handle image OCR and live status events [4].
The that combines web and mobile clients, an application API, a text model operates on selected masked text, while the record path classifier, evidence processing, and officer review.
This paper reports retains complaint data.
Those paths have different privacy a group-aware evaluation on 1,950 synthetic narratives across 13 implications and must be evaluated separately.
canonical classes.
The dataset contains 130 scenario families, with eight families per class in development and two held out; the This paper replaces the earlier 257-sample snapshot and 39-1,560/390 split has no shared groups.
The v3 experiment summary item holdout with the v3 scenario-family split.
It documents reports 88.97% held-out accuracy (347/390) and 0.8896 macro-F1.
the service topology, label and partition design, model Five-fold GroupKFold comparison selected Logistic Regression configuration, group-aware benchmark, held-out and (C=2.0) for its native class-probability output and threshold-based challenge results, and limitations.
Reported metrics are rejection, although LinearSVC attained higher cross-validation specific to synthetic or programmatically authored data; they accuracy.
A 215-item programmatically authored challenge set do not establish deployment safety, law-enforcement produced 98.46% raw and 97.44% gated accuracy on its 195 in-domain items.
These results do not establish performance on real acceptance, or generalization to citizen complaints.
complaints. The challenge set has not been validated by sworn law-II.
RELATED WORK enforcement desk officers.
PrivacyShield masks selected text only for model input; original complaint and OCR data may remain stored.
Crime-report classification and similarity have been studied The classifier is an aid to human triage, not a decision-maker.
as tools for organizing reports [1].
Riekert et al.
compare Index Terms— cybercrime complaints, group-aware evaluation, text support-vector and naive Bayes text baselines
across public classification, TF-IDF, Logistic Regression, PII masking, out-of-datasets [5], while Rani et al.
examine transformer-based distribution rejection.
classification for Hinglish cybercrime complaints using a separate corpus [6].
Differences in task, language, and data I.
INTRODUCTION prevent direct score comparisons with CyberGuard.
Cybercrime complaint systems must organize narrative NIST guidance treats PII protection as context-dependent reports without treating a model output as verification of an and distinguishes masking from risk-assessed de-identification allegation.
Prior work has applied text classification and report [2], [3].
CyberGuard therefore treats the classifier input as one similarity to crime-report analysis [1].
CyberGuard addresses a data path, not as a proxy for storage, access, retention, narrower engineering task: suggesting one of 13 categories evidence handling, or disclosure controls.

from synthetic English narratives while preserving an officer-led review process.
III. PROBLEM DEFINITION AND OBJECTIVES The application connects web and mobile clients to an A complaint contains narrative text and structured fields.
Express API, MongoDB, and a Flask classifier; separate The intended workflow validates the submission, obtains a category suggestion, stores the case and evidence references, IV.
SYSTEM ARCHITECTURE AND END-TO-END and supports review and status updates by authorized WORKFLOW personnel.
The classifier predicts within a closed label set; it Fig.
1 summarizes the inspected service topology.
React and does not determine whether a report is true, establish liability, React Native clients submit complaint data to the Express API, or assess legal sufficiency.
which validates requests, applies selected access checks, stores The classifier output is distinct from the final stored case records, and calls the Flask classifier [4].
The classifier category.
The application can accept a user-selected category receives masked text and returns predicted probabilities, while and applies separate controller and schema rules [4].
the backend also provides a local keyword fallback if the Flask Evaluation of the model therefore concerns only the text service is unavailable.
classifier unless an end-to-end application test is explicitly Officer and administrator actions can update case status and reported.
append timeline events.
Socket.IO emits real-time events to Authentication and role checks restrict selected operations, user or administration rooms and supports live status updates while evidence access and complaint views follow separate [4].
Public lookup routes permit status queries using complaint authorization paths [4].
Public tracking, search excerpts, stored identifiers, requiring care that sensitive details are not exposed OCR text, and logs require independent privacy review.
outside authorized sessions.
Model-input masking is not a system-wide privacy guarantee, and the present offline evaluation does not test route-level authorization or deployment security.
Fig. 1.
System architecture and service topology of CyberGuard.
Fig. 2 presents complaint intake and triage.
The application transaction validity or account ownership.
OCR accuracy, file masks supported identifiers before forwarding text to the security, and chain of custody remain unvalidated.
classifier. On request failure, keyword rules assign a heuristic Severity and priority are produced by application rules suggestion.
Schema and controller rules can alter categories or rather than by the model.
Flags such as immediate_action adjust priority, and the original narrative is retained for officer trigger high-priority status, and safety categories are review.
prioritized by schema hooks [4].
The interaction between rule The repository uses Tesseract.js for English-language image overrides and model suggestions requires explicit system-level OCR [4], [7], [8].
OCR text and regex-extracted identifiers tests.
(phone numbers, UPI handles, account numbers) are saved The saved category can differ from the classifier prediction with the complaint record [4].
Regex matches do not verify because user selection or administrative edits can override model outputs [4].
The evaluation in this paper assesses model performance independently of application-level overrides.
Fig. 2.
Complaint processing workflow and triage flowchart, including classification, fallback, evidence OCR, persistence, and human review.
A. Implemented Components and Feature Boundaries B.
Repository Technology Stack and Service Topology TABLE I TABLE II IMPLEMENTED COMPONENTS AND FEATURE

BOUNDARIES REPOSITORY TECHNOLOGY STACK AND SERVICE TOPOLOGY Component Implemented role and evidence boundary Layer Repository technology and service role Web/mobile clients React and React Native clients support Web/mobile React and React Native/Expo clients for complaint intake, tracking, and review; no intake, tracking, and review [4].
usability study is reported [4].
Application/API Node.js, Express, REST, JWT Application/API Node.js and Express validate requests, apply authentication, selected role checks, and selected JWT and role checks, and connect service integration [4].
application services [4].
Persistence MongoDB/Mongoose complaint records, Persistence MongoDB/Mongoose stores complaint predictions, timelines, evidence references, fields, category and prediction data, and OCR data [4].
timelines, evidence references, and OCR text [4].
Classification Python, Flask, scikit-learn, word TF-IDF (1– 2 grams), and Logistic Regression (C=2.0) PrivacyShield Regular-expression masking changes [4].
selected model input; the source record may remain stored.
Coverage and error rates are Evidence/realtime Local file handling, Tesseract.js image OCR, unmeasured [4].
and Socket.IO event delivery [4], [7].
Classifier service Flask serves word-level TF-IDF and Logistic Regression.
This paper evaluates the v3 configuration offline [4].
V. IMPLEMENTATION AND MACHINE-LEARNING METHODOLOGY OOD and triage A probability threshold supports rejection for review; application fallback, user choice, and A.
Dataset and Group-Aware Partitioning priority rules remain separate [4].
The v3 dataset contains 1,950 synthetic complaint Evidence and OCR Local evidence handling and English image narratives, balanced across 13 canonical classes with 150 OCR are implemented; accuracy, integrity, examples per class.
The categories are: child_exploitation, and production security are not established crypto_fraud, cyberbullying, data_breach, financial_fraud, [4], [7].
hacking, identity_theft, network_attacks, online_fraud, other, Status and events Timeline and Socket.IO functions support phishing, ransomware, and women_child_safety [4].
updates; delivery and concurrent operation were not evaluated [4].
Fig. 3.
Class distribution of the 1,950-sample synthetic dataset across 13 canonical classes.
Scenario-family membership defines the partition.
The The selected pipeline uses word-level TF-IDF with unigram development set contains 1,560 samples across 104 families and bigram features, sublinear term-frequency scaling, and a (eight families per class), while the held-out test set contains 10,000-feature ceiling [9].
The classifier is multinomial 390 samples across 26 unseen families (two families per Logistic Regression with C=2.0 and balanced class weighting, class).
The partition was verified to have zero group overlap.
fit using the L-BFGS solver [10].
Table III summarizes reconciliation with earlier project Model selection used five-fold GroupKFold cross-claims.
validation on the 1,560-sample development set, with TF-IDF TABLE III fit strictly inside each training fold to prevent group leakage.
RECONCILIATION OF DOCUMENT AND REPOSITORY CLAIMS LinearSVC achieved the highest cross-validation accuracy (81.79%), but Logistic Regression (79.88%) was chosen for its Claim V3 evidence and disposition native class probabilities and threshold-based rejection Dataset snapshot The earlier 257-sample snapshot is capability.
superseded by the 1,950-row v3 dataset; 150 The benchmark reports inference latencies of 1.1 ms for samples per class.
Resolved for this LinearSVC, 1.6 ms for Logistic Regression, and 0.4 ms for manuscript [4].
MultinomialNB on synthetic text [4].
These measurements Canonical labels Thirteen canonical classifier labels are used exclude network transport and preprocessing overhead.
throughout; 150 examples per label.

TABLE IV Resolved [4].
CANDIDATE MODEL BENCHMARK AND FINAL EVALUATION Partition 1,560 development and 390 held-out OUTPUT samples, split by scenario family; zero group Evaluation Stage & Empirical Metrics & Operational Role overlap.
Resolved [4].
Configuration Held-out accuracy The v3 experiment summary reports 88.97% LinearSVC (C=2.0) [5-Accuracy: 81.79% ± 2.29%; Macro-F1: (347/390), adopted here.
A separate report Fold Group CV] 0.7869; Latency: 1.1 ms.
Highest CV lists 89.74%; source discrepancy disclosed accuracy; requires Platt calibration for [4].
threshold gating [4].
Challenge results The main summary reports 97.44% gated in-Logistic Regression Accuracy: 79.88% ± 3.15%; Macro-F1: domain accuracy; another artifact reports (C=2.0) [Selected] 0.7600; Latency: 1.6 ms.
Selected for native 96.92%.
Exact prediction artifacts are needed class probabilities and threshold-based OOD for reconciliation [4].
rejection at tau=0.20 [4].
B. Model Specification and Candidate Benchmark Evaluation Stage & Empirical Metrics & Operational Role Configuration Table IV reports candidate comparisons on the development MultinomialNB Accuracy: 76.92% ± 2.18%; Macro-F1: set alongside the final held-out and challenge benchmark (alpha=0.1) [5-Fold 0.7317; Latency: 0.4 ms.
Fastest inference, outputs.
The selected model balances group-disjoint Group CV] but lowest accuracy and severe probability generalization, probability support, and operational distortion [4].
integration. Held-Out Test Partition Accuracy: 89.74% (350/390); Macro-F1: C.
PrivacyShield and Prediction Pipeline (N=390 Unseen) 0.8975; Precision: 0.9106; Recall: 0.8974.
PrivacyShield replaces supported regular-expression Clean subset: 89.66%; 5% noisy subset: 92.31% [4].
matches in selected narratives with indexed tokens (e.g., EMAIL_1, PHONE_1) prior to classification [4].
This reduces Challenge Benchmark Raw Accuracy: 97.95% (191/195); Gated exposure of direct identifiers during model inference.
(N=215 Items) Accuracy: 96.92% (189/195); In-Domain FRR: 1.03%; Benign OOD Safe Rejection: 95.0% (19/20) [4].
Fig. 4.
PrivacyShield model-input PII masking pipeline, with original complaint storage shown separately.
Masking coverage and false-match rates have not been (classifier.pkl) evaluated across all 390 held-out test samples.
established for free-form narratives or Indian vernacular terms.
The model correctly classifies 350 of 390 test narratives, Structured complaint fields and uploaded evidence are stored achieving 89.74% accuracy and 0.8975 macro-F1 across the in their submitted form.
Model-input masking does not 13 canonical classes.
An earlier preliminary solver checkpoint provide system-wide privacy or compliance with data-recorded 88.97% accuracy (347/390) and 0.8896 macro-F1; protection regulations [2], [3].
Table III documents this reconciliation.
Both evaluations share the identical 1,560/390 group-disjoint scenario-family split VI.
HELD-OUT EVALUATION with zero template overlap, and the production artifact After candidate selection, the Logistic Regression pipeline confirms the 350/390 (89.74%) benchmark [4].
was evaluated on the 390-sample held-out partition.
The The threshold tau = 0.20 was evaluated as an out-of-pipeline was fit on the 1,560 development samples and distribution rejection rule.
Because the classifier uses standard evaluated on the 26 unseen scenario families.
softmax without Platt scaling or isotonic calibration, softmax The empirical confusion matrix in Fig.
5 reports the exact outputs reflect relative class scores rather than calibrated predictions of the serialized production model artifact posterior probabilities under real-world distribution shift.
Fig. 5.
Confusion matrix on the 390-sample held-out test partition.

The primary v3 summary reports perfect F1 for phishing VII.
GROUP-AWARE COMPARISON AND and ransomware, and F1 values exceeding 0.90 for CHALLENGE EVALUATION
crypto_fraud, identity_theft, and financial_fraud.
Lower recall was observed for women_child_safety (confused with Figure 6 compares the three candidate classifiers using five-
cyberbullying and child_exploitation) and hacking (confused fold GroupKFold results on the development partition.
with data_breach).
These error patterns reflect lexical overlap LinearSVC achieved the highest mean accuracy (81.79% ± among related offense
types.
2.29%) and macro-F1 (0.7869), but its decision-function margins do not provide native probabilities.
Logistic The held-out evaluation used scenario families rather than Regression (79.88% ± 3.15%, macro-F1 0.7600) was
preferred random record-level splits, providing a more demanding test because its multinomial softmax outputs enable threshold-
than earlier fixed-split evaluations.
However, all samples based triage gating without an auxiliary calibration layer.
remain synthetic, and performance on authentic citizen Multinomial Naive Bayes showed lower accuracy (76.92% ± complaints
cannot be inferred from these results.
2.18%, macro-F1 0.7317).
Fig. 6.
Candidate classifier comparison across five-fold GroupKFold cross-validation.
The independent challenge collection contains 215 96.92% gated accuracy (189/195), and 95.0% safe rejection of
programmatically authored items: 195 in-domain complaints benign queries (one false acceptance) [4].
(15 per class across all 13 categories) and 20 benign non-cyber Challenge artifacts also differ on the gated in-domain result
queries.
The main evaluation summary reports 98.46% raw and benign-query false-acceptance rate.
These discrepancies accuracy (192/195) and 97.44% gated accuracy (190/195) with are documented in Table III.
The challenge benchmark was zero false acceptances on the 20 benign queries.
The serialized authored programmatically by the research team and has not model artifact yielded 97.95% raw accuracy
(191/195), been reviewed or validated by sworn law-enforcement personnel.
TABLE V LIMITATIONS AND REQUIRED VALIDATION Limitation or risk Observed boundary and required validation Synthetic
data All 1,950 training and test narratives are synthetic.
Validate on independently sourced, privacy -reviewed cases with documented human annotation [4].
Challenge provenance The 215-item set was programmatically authored and has not been validated by sworn law-enforcement
desk officers [4].
Probability and OOD gating Soft probability outputs and tau=0.20 do not establish calibration or robust rejection under distribution
shift.
Assess calibration and threshold harms prospectively [4].
Human override Maintain desk review for all predictions, rejections, disagreements, and service failures.
Do not automate complaint rejection or legal decisions.
Privacy and evidence Test masking, storage, OCR, logs, access, retention, deletion, and evidence integrity as separate controls
[2]–[4], [7].
Artifact consistency Resolve conflicting held-out and challenge metrics; release the exact split, model, predictions, and code
revision with each reported result [4].
All model-development data are synthetic.
A future study operational risk assessment against missed referrals and false needs independently collected, privacy-reviewed
complaints alarms [4].
with documented human annotation to establish real-world Human desk review must remain available for every generalization
[4].
suggestion, rejection, disagreement, and fallback event.
The Logistic Regression outputs are soft probabilities.
Automated classification must not determine complaint Selection of tau = 0.20 requires empirical calibration and admissibility or

replace officer judgment.
Privacy validation must cover stored narratives, [1] C.-H.
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[Online]. Available: https://scikit- (350/390) and 0.8975 macro-F1 on group-disjoint scenario
learn.org/1.6/modules/generated/sklearn.feature_extraction.text.Tfid families (reconciling the 88.97% / 347-of-390 preliminary
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Classification metrics,” GroupKFold cross-validation accuracy (macro-F1: 0.7600) on scikit-learn 1.6 documentation.
[Online]. Available: https://scikit-the development partition.
The deployed serialized model learn.org/1.6/modules/model_evaluation.html#classification-metrics artifact achieves 89.74% held-out accuracy and 96.92% operational gated accuracy on an independent 215-item challenge benchmark, with 95.0% safe rejection of benign queries.
Logistic Regression was selected over higher-accuracy LinearSVC because its multinomial softmax outputs provide native class-probability scores directly usable for threshold-based out-of-distribution rejection, without requiring an auxiliary scaling layer, though empirical probability calibration under real-world distribution shift was not demonstrated.
Architecturally, PrivacyShield masks supported PII in text forwarded to the machine-learning pipeline, while raw complaint narratives, structured fields, and uploaded evidence files are stored separately as prototype records in MongoDB for triage inspection.
Human-in-the-loop desk officer verification, production security hardening, and prospective field trials remain indispensable prerequisites prior to operational municipal deployment.
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